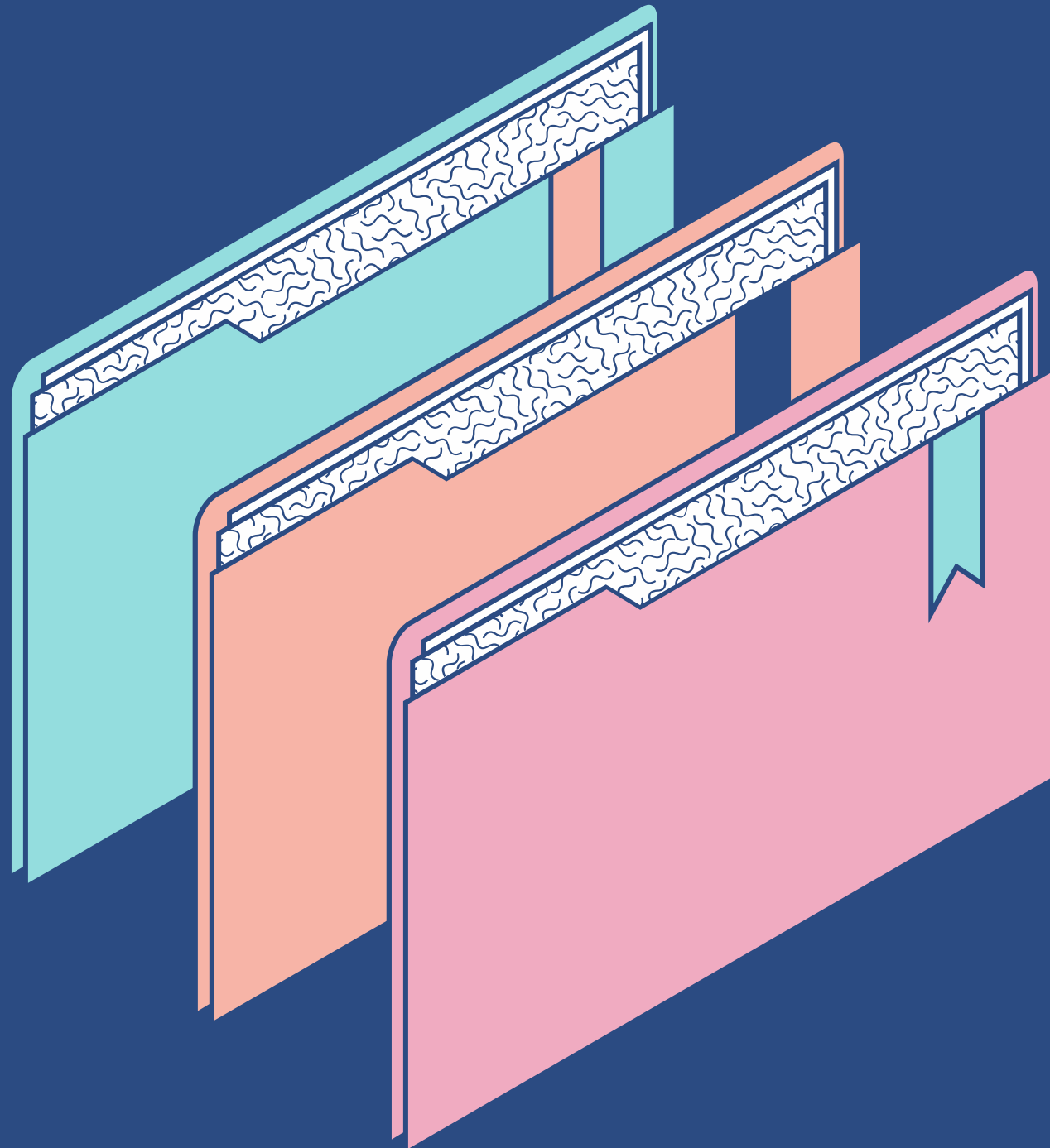




A QUICK INFORMATION GUIDE

Data Warehousing

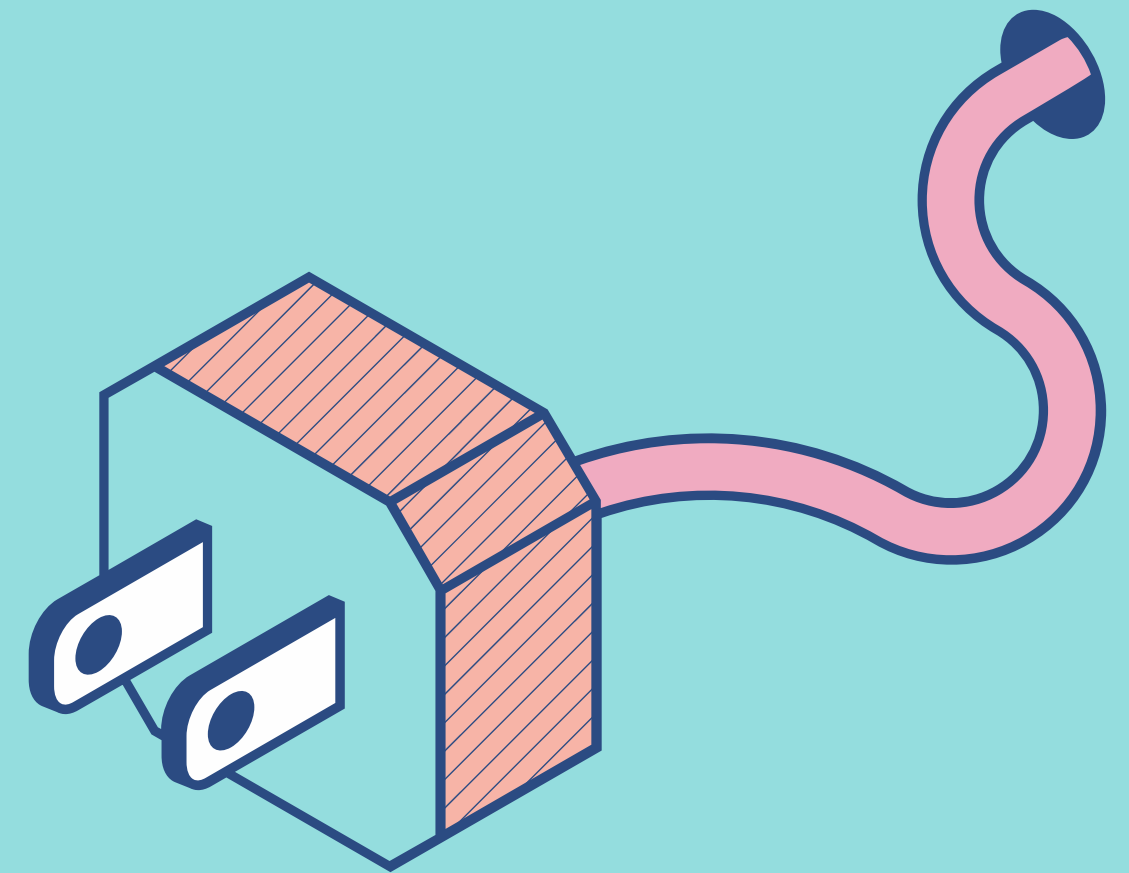
Data Warehouse

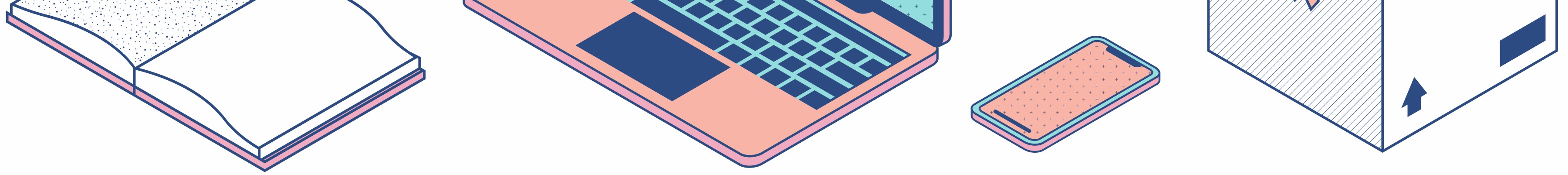


- **storage of information** over time by a business or other organization.
- new data is periodically added by people in various key departments
- becomes a library of historical data that can be retrieved and analyzed for decision-making
- key factors include defining critical information to the organization and identifying its sources
- designed to supply real-time information. A data warehouse is designed as an archive of historical information.

Use in Data Mining

- Businesses warehouse data primarily for data mining
 - looking for patterns of information that will help them improve their business processes.
- A good data warehousing system makes it easier for different departments within a company to access each other's data
 - marketing team can assess the sales team's data in order to make decisions about how to adjust their sales campaigns.





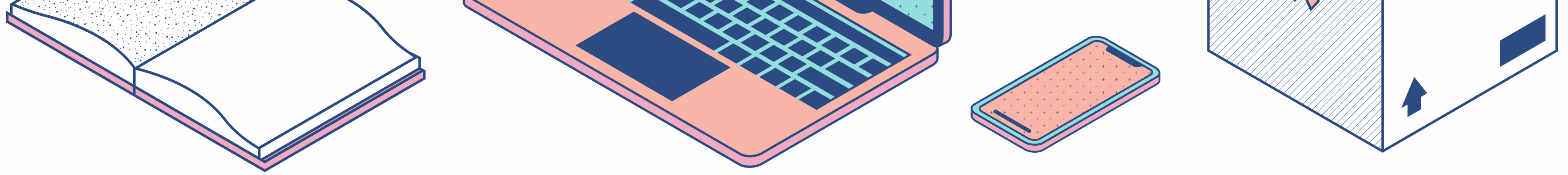
Data Warehouse

vs.

Database

- A data warehouse is programmed to aggregate structured data over time.

- A database is a transactional system that monitors and updates real-time data in order to have only the most recent data available.



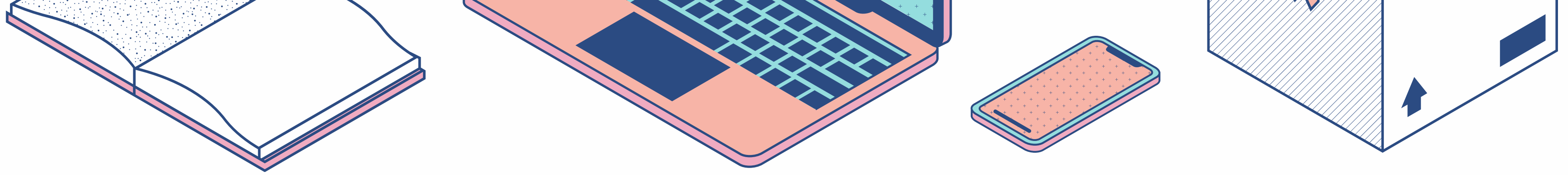
Data Warehouse

vs.

Data Lake

- A data warehouse holds refined data that has been filtered to be used for a specific purpose
- most often used by business professionals
- more structured and any changes are more costly

- a data lake holds raw data of which the goal has not yet been determined
- primarily used by data scientists
- more easily accessible and easier to update



Data Warehouse

vs.

Data Mart

- A data warehouse holds refined data that has been filtered to be used for a specific purpose
- most often used by business professionals
- more structured and any changes are more costly

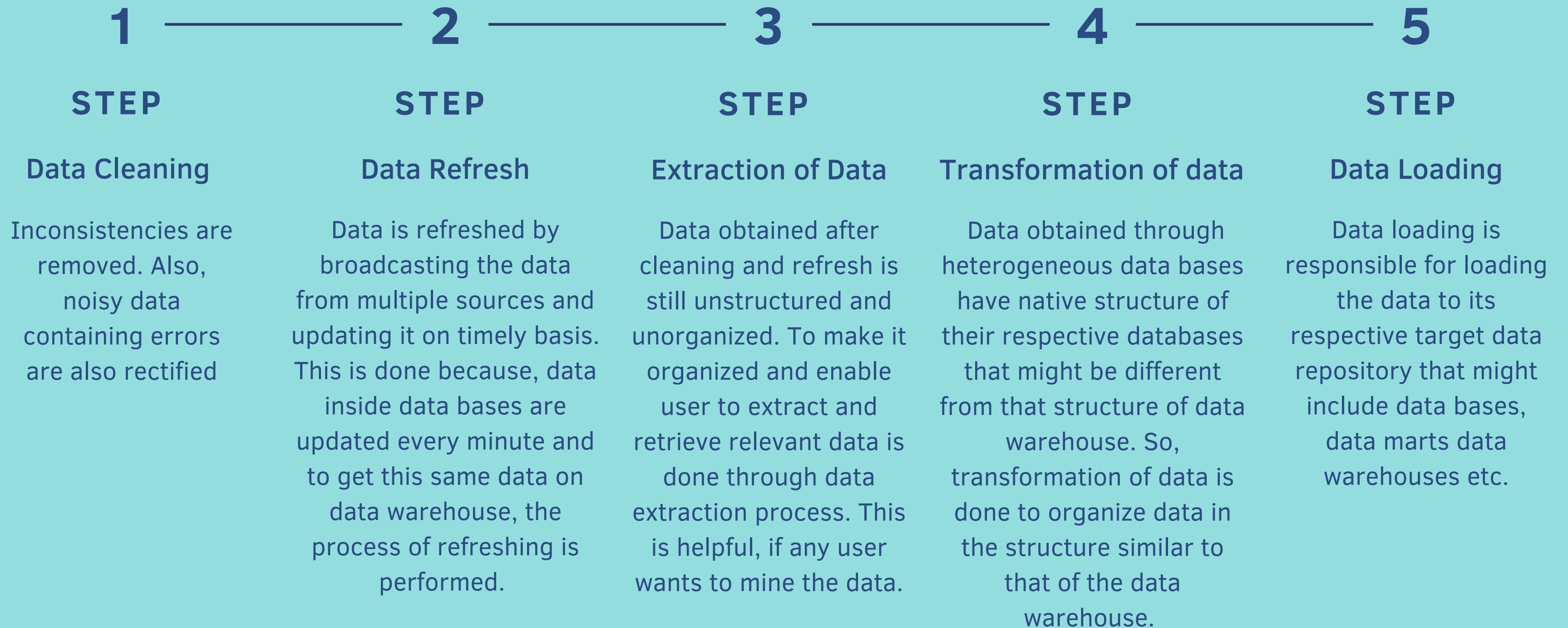
- A data mart is just a smaller (subset) and faster version of a data warehouse
- collects data from a small number of sources and focuses on one subject area
- focus on one area for analytical purposes, such as a specific department within an organization
- used to help make business decisions by helping with analysis and reporting



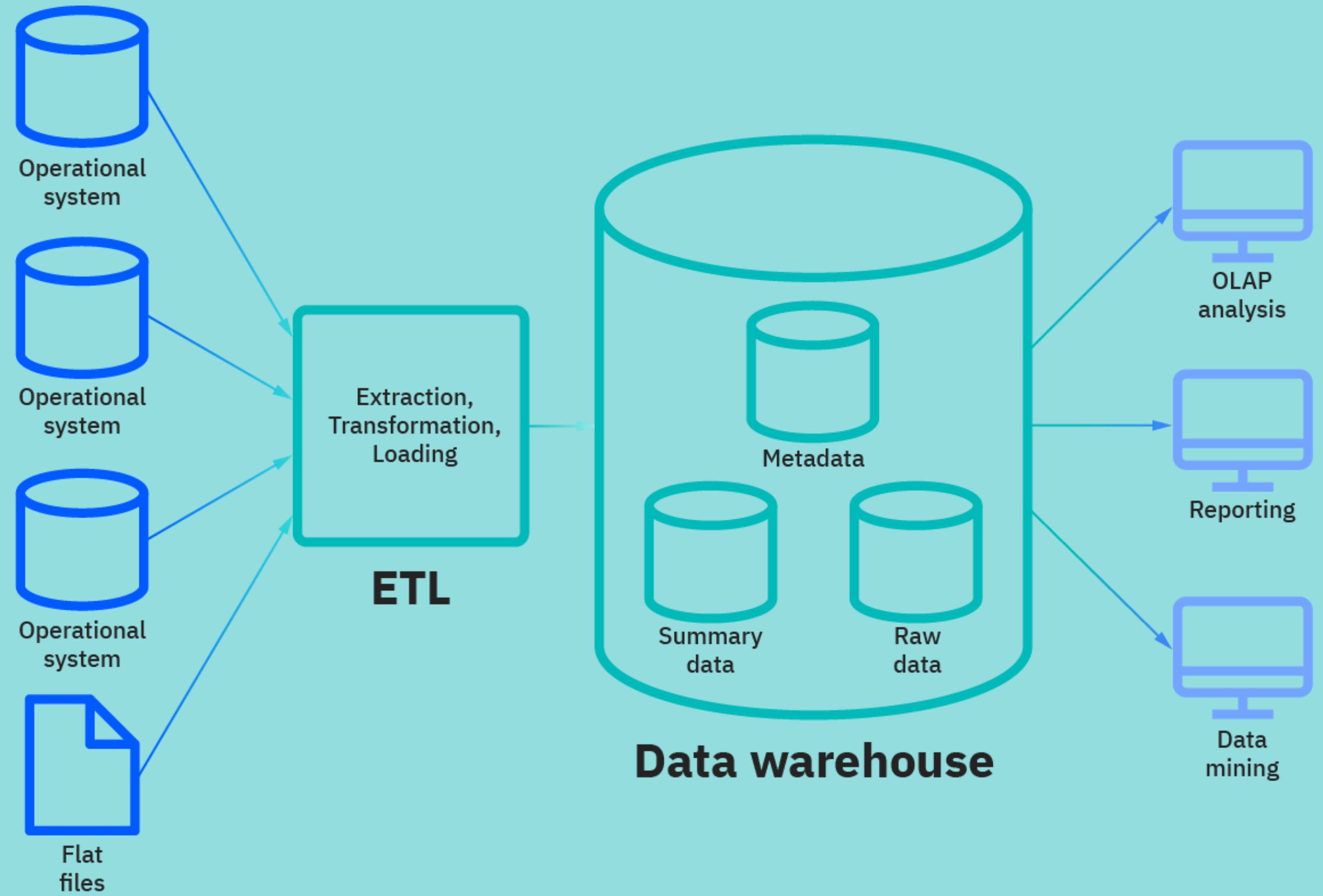
Data Warehouse Operations

- Any data warehouse will consist of random data which will surely be in unstructured manner with a lot of unwanted and dirty data.
- To make this data structured and noise free, dirty data needs to be removed, converting data into useful information and can be achieved using certain data warehouse operations.

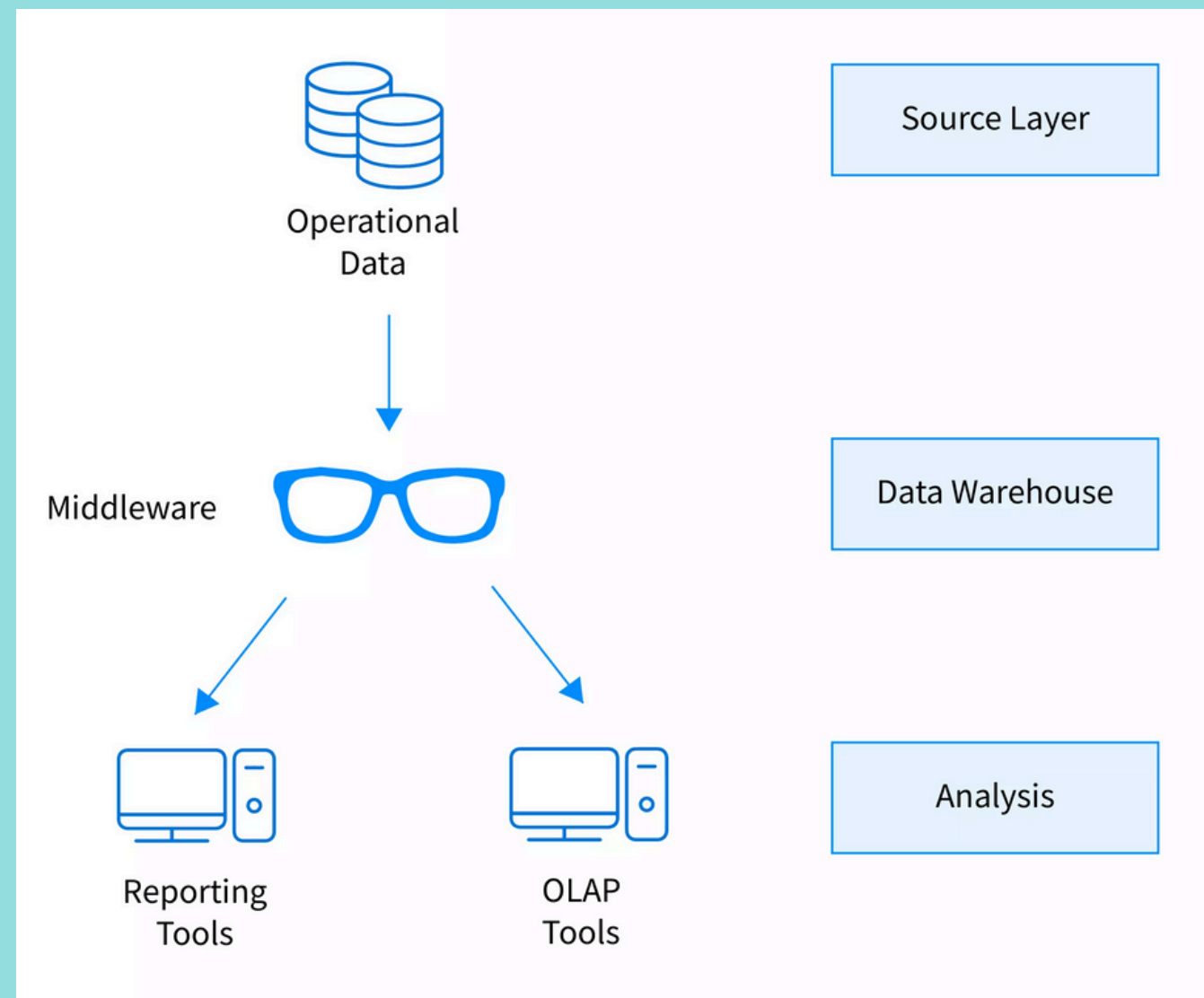
Data Warehouse Operations



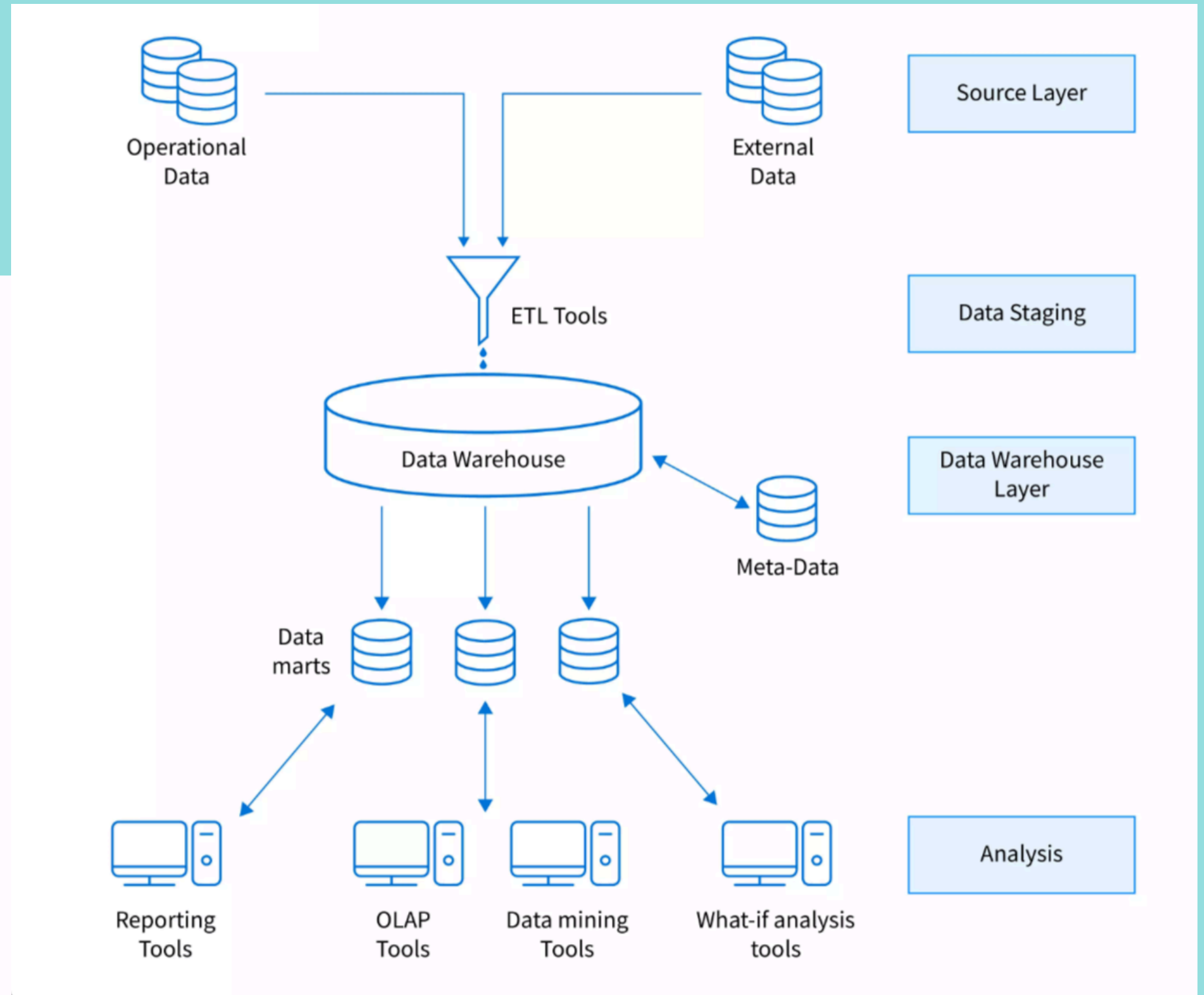
Data Warehouse Operations



Data Warehouse Architecture



Single-tier Architecture

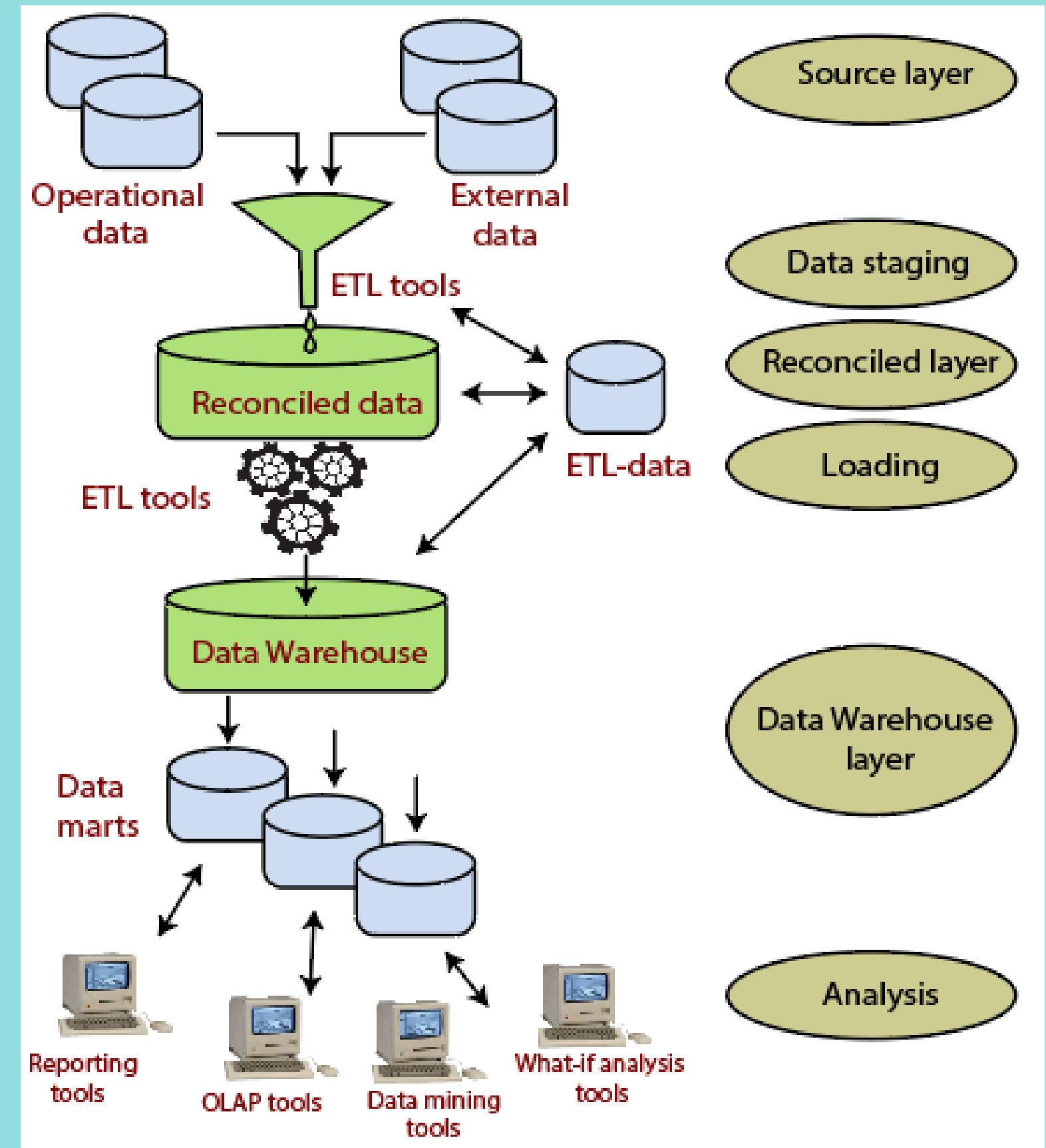


Two-tier Architecture

Data Warehouse Architecture

Three-Tier Architecture

- has a top, middle, and bottom tier (source layer, the reconciled layer, and the data warehouse layer)
- suited for systems with long life cycles
- an extra layer of review and analysis of the data is completed when changing data to ensure there have been no errors



Layers of data warehouse architecture

Source layer	Staging layer	Warehouse layer	Consumption layer
• feed data into the warehouse • point-of-sale, marketing automation, CRM, or ERP systems • has a specific data format and may require a different data capture method based on that data format.	• landing area for data from the source • ingest data from the SOR without applying business logic or transformations • not used in production data analysis • data in the staging area has yet to be cleansed, standardized, modeled, governed, and verified.	• where all of the data is stored • warehouse data is now subject-oriented, integrated, time-variant, and non-volatile • will have the physical schemas, tables, views, stored procedures, and functions needed to access the warehouse-modeled data	• analytics layer, where you model data for consumption using analytics tools • like ThoughtSpot, data analysts, data scientists, and business users

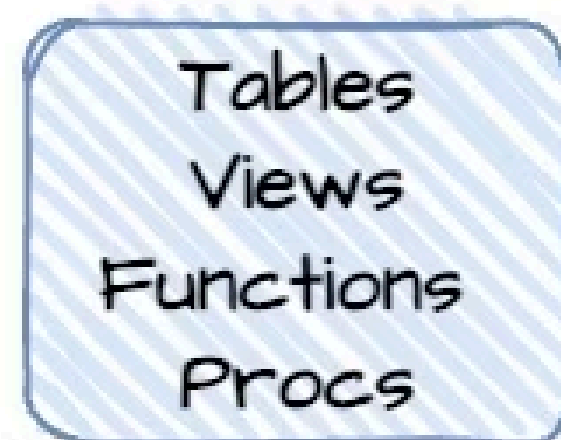
Source
Layer



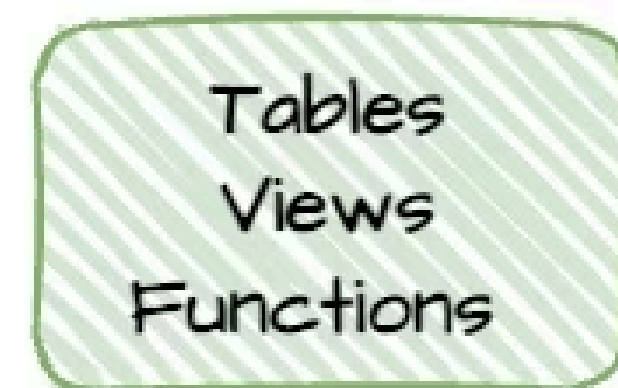
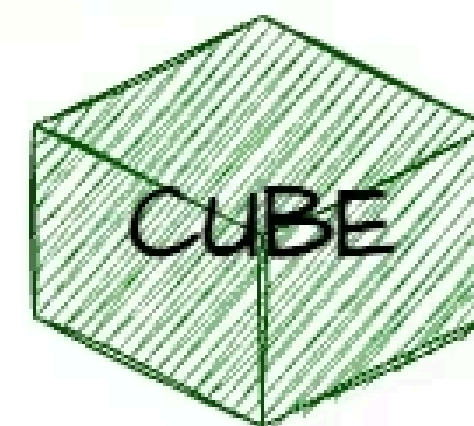
Staging
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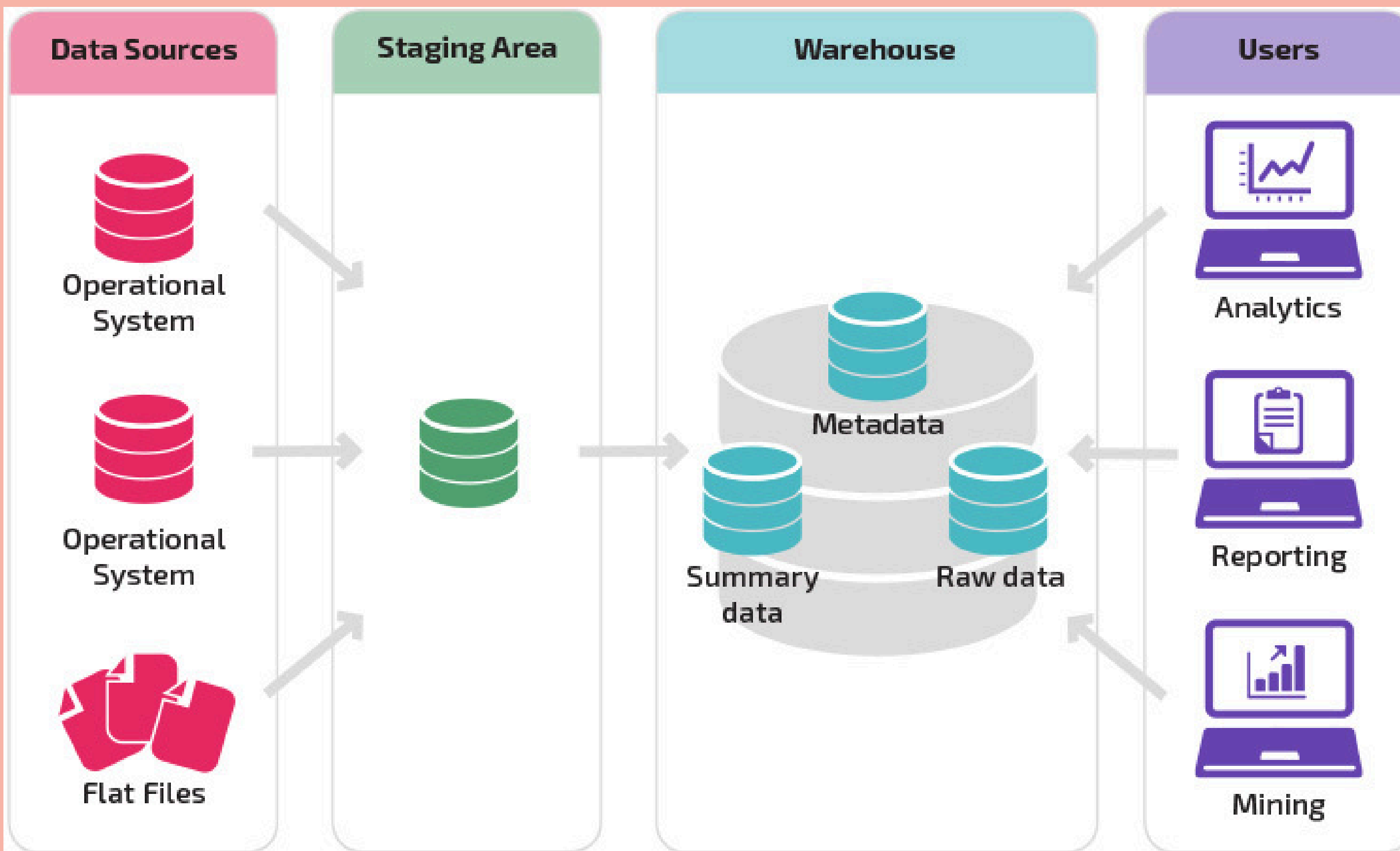


Warehouse
Layer



Consumption
Layer





Do you have any questions?

If none, then thank you for listening.

